

REVIEW OPEN ACCESS

Smart Hydrogels for Sustainable Agriculture

Jungjoon Park | Weixin Guan | Guihua Yu 

Materials Science and Engineering Program and Walker Department of Mechanical Engineering, The University of Texas at Austin, Austin, Texas, USA

Correspondence: Guihua Yu (ghyu@austin.utexas.edu)**Received:** 8 December 2024 | **Revised:** 16 February 2025 | **Accepted:** 17 March 2025**Funding:** This work was supported by Welch Foundation, F-1861.**Keywords:** agriculture | agrochemical | controlled delivery | hydrogel | sensor

ABSTRACT

The growing global population, coupled with increasing food demand and water scarcity, has intensified the need for advancements in modern agriculture. As an emerging class of materials featured by intensively tunable properties, smart hydrogels offer innovative solutions to challenges associated with conventional agricultural practices, such as excessive agrochemical and water use and inefficiencies that contribute to environmental degradation. Additionally, hydrogel-based sensors can monitor environmental conditions and crop health, enabling precise adjustments to optimize growth and resource use. By serving as platforms for the slow and controlled delivery of agrochemicals and smart sensors, hydrogel systems can enhance resource efficiency, reduce labor demands, and improve crop yields in an environmentally sustainable manner. This Perspective article summarizes recent advancements in hydrogel-based materials, highlights existing challenges, and proposes potential research directions, with a focus on developing advanced hydrogel systems to transform agricultural practices.

1 | Introduction

The global population is projected to surpass 9.7 billion by 2050 [1], necessitating agricultural systems to address a significant increase in food demand. Quantitative scenario studies indicate that global food demand is expected to rise by 35% to 56% between 2010 and 2050 under a range of socio-economic and climate scenarios [2]. This demand is driven by population growth and shifting dietary preferences, particularly in emerging economies. However, these needs coincide with an impending water crisis, with nearly half of the global population predicted to experience severe water scarcity by 2050 [3, 4]. Agriculture, which currently consumes over 70% of global freshwater resources, will be forced to adapt to diminishing water availability while simultaneously increasing productivity.

Traditional agricultural practices have historically supported food production growth, but their heavy reliance on synthetic fertilizers, pesticides, and herbicides has imposed severe environmental costs. Excessive chemical inputs contribute to

nutrient leaching, groundwater contamination, and the accumulation of toxic residues, reducing arable land quality and threatening biodiversity (Figure 1). Nutrient use efficiency (NUE) of chemical fertilizers is alarmingly low, at <30%, while modern agricultural pest management loses ~90% of applied pesticides, resulting in significant ecological harm [5, 6].

Amid these challenges, smart hydrogels represent a promising approach for sustainable agriculture. These polymeric materials, with water retention from hydrophilic networks and controlled release from polymer matrices, address critical issues of water scarcity and inefficient agrochemical delivery. Hydrogels can absorb and retain large volumes of water, gradually releasing moisture to the soil and reducing irrigation requirements. Additionally, their capacity for controlled agrochemical delivery minimizes nutrient losses, improves pesticide efficiency, and aligns with plant growth cycles. Certain conductive hydrogels also function as biosensors, providing real-time data on soil conditions such as moisture, pH, and nutrient levels, which enhance the precision and efficacy of agricultural interventions.

This is an open access article under the terms of the [Creative Commons Attribution](https://creativecommons.org/licenses/by/4.0/) License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

© 2025 The Author(s). *EcoMat* published by The Hong Kong Polytechnic University and John Wiley & Sons Australia, Ltd.

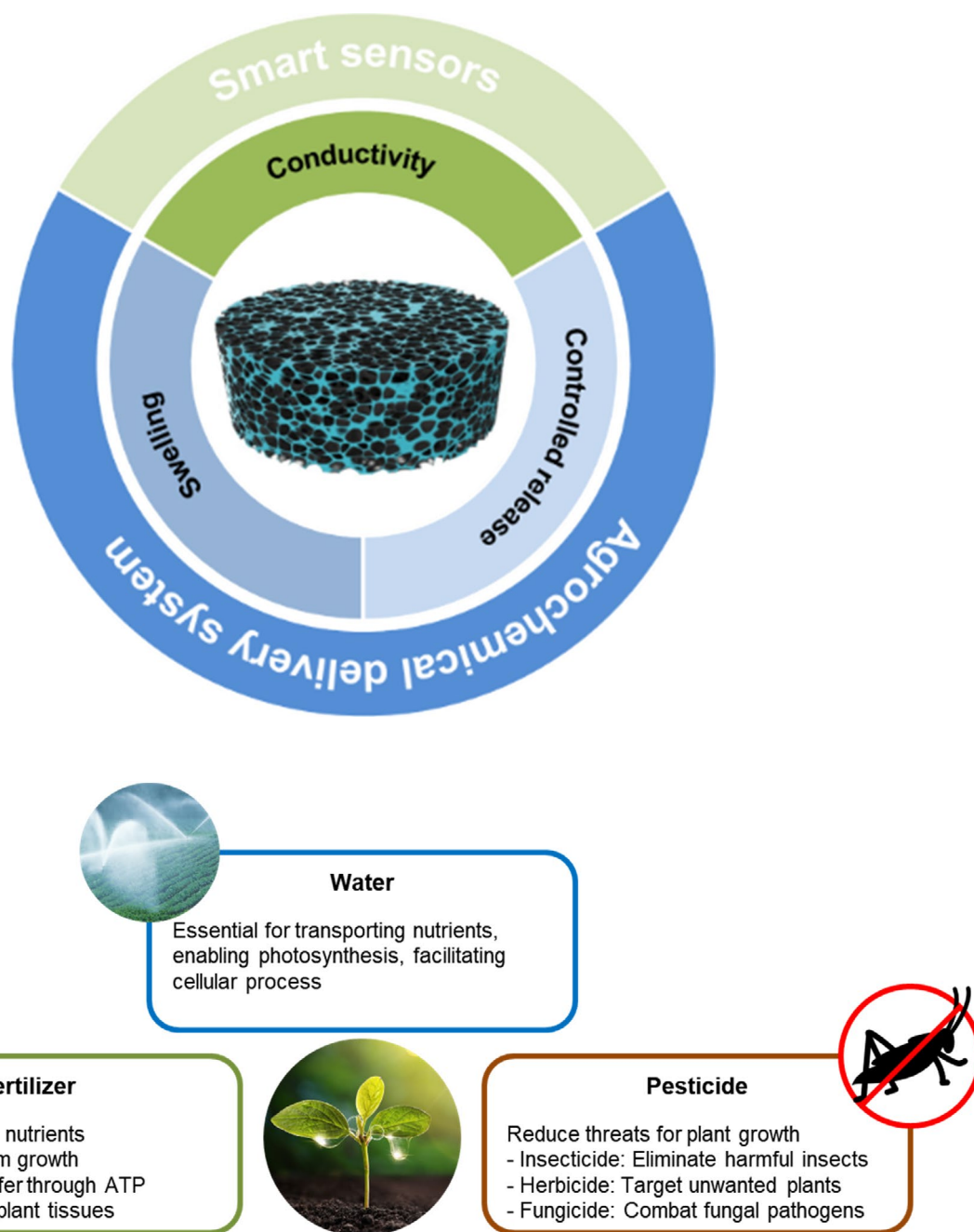


FIGURE 1 | Essential inputs (water, fertilizer, pesticide) for plant growth cycle.

By integrating water retention, nutrient management, and sensing functionalities, smart hydrogels hold the potential to tackle the intertwined challenges of food security, resource scarcity, and environmental degradation. This review explores the latest advancements in hydrogel technologies, highlighting their multifaceted applications in agrochemical delivery and smart sensor systems, with the potential to revolutionize agricultural practices and achieve the dual goals of sustainability and global food security.

2 | Key Factors Driving Plant Growth

Plant growth and germination are directly influenced by water, nutrients, and protection from biotic stressors throughout the

growth cycle. These factors are essential for successful agricultural practices, ensuring that plants receive the resources they need to develop and resist environmental threats. By understanding and optimizing these critical factors, agricultural practices can be adjusted to enhance crop productivity and resilience in varying climate conditions and limited resources.

Water is foundational for all physiological processes in plants, serving as a solvent, a transport medium, and a vital component in photosynthesis. It is essential for plant structure and function through its role in maintaining cellular turgor pressure, which ensures that cells remain rigid and capable of supporting growth. Without adequate hydration, photosynthesis becomes difficult, hindering growth and reducing productivity. Efficient water management ensures that plants receive the appropriate

amount of hydration at each stage of development. During germination, water activates enzymes and softens the seed coat, allowing the radicle to emerge. As plants grow, water facilitates nutrient absorption from the soil, as nutrients dissolve in water before being transported to various parts of the plant via the xylem. Overwatering, however, can lead to waterlogging, which suffocates roots and inhibits nutrient uptake, while under-watering causes drought stress.

Plants require a combination of macronutrients and micronutrients to sustain growth and complete their life cycle. Among these, nitrogen (N), phosphorus (P), and potassium (K) are the most critical and are often referred to as the primary macronutrients. Fertilizers rich in nitrogen, such as urea or ammonium nitrate, are typically applied during the vegetative stage to promote leaf and stem development. Nitrogen is essential for chlorophyll synthesis, as it directly supports photosynthesis. It promotes vegetative growth by aiding in the production of amino acids, the building blocks of proteins. A lack of nitrogen causes plants to appear pale, with yellowing leaves (chlorosis) and stunted growth, as they cannot synthesize sufficient chlorophyll. Proper nitrogen application ensures healthy foliage and robust early development. Phosphorus is essential for energy transfer as a key component of adenosine triphosphate, which drives cellular activities. It plays a crucial role in root development, helping plants establish a strong foundation for nutrient and water uptake. Phosphorus also supports flowering and fruit production during the later stages of plant growth. Insufficient phosphorus results in poor root systems, delayed maturity, and reduced reproductive success. Potassium is critical for regulating water balance within plant tissues, enhancing drought resistance and tolerance to environmental stresses such as salinity. It strengthens cell walls, improving tissue integrity and reducing susceptibility to pests and diseases. Crops with adequate potassium exhibit stronger stems, better water use efficiency, and enhanced stress tolerance. Deficiency in potassium can lead to weak stems, poor stress resistance, and reduced productivity.

Pesticides are critical in agriculture, protecting plants from pests, weeds, and pathogens that threaten their growth and yield. They include insecticides, herbicides, and fungicides. Insecticides secure crops from insect pests such as aphids, caterpillars, and borers, which feed on leaves, stems, or roots and can cause significant damage. Herbicides are essential for eliminating weeds that compete with crops for light, water, and nutrients, particularly during the early growth stages when plants are most vulnerable. Selective herbicides, such as those used in cornfields, effectively target weeds without harming crops, ensuring optimal resource availability and maximizing the benefits of applied water and fertilizers. Fungicides protect crops from pathogenic fungi, including those that cause rust, mildew, or blight, which can attack plants at any stage of growth. They are especially vital in humid climates, where fungal infections spread rapidly.

3 | Hydrogels: Core Properties, Agricultural Attributes, and Smart Applications

Hydrogels are three-dimensional (3D) crosslinked polymeric networks distinguished by their ability to absorb and retain

significant amounts of water, a property known as swelling. This swelling behavior is fundamental to their use in agriculture, facilitating water retention, controlled agrochemical delivery, and soil enhancement [7]. The swelling behavior is governed by both the functionality of the polymer chains and the network architecture. Hydrophilic functional groups, such as hydroxyl ($-\text{OH}$), carboxyl ($-\text{COOH}$), amide ($-\text{CONH}_2$), and ether groups ($-\text{CH}_2\text{OCH}_2-$), interact with water molecules through hydrogen bonding, enhancing the hydrogel's water uptake [8]. The nature of the molecular interaction of water and these functional groups can be intricately linked to hydration number, which represents the number of molecules associated with a solute [9]. Ionic groups, such as carboxylates and sulfonates, further contribute to swelling by generating electrostatic repulsion and osmotic pressure gradients within the hydrogel network [10], which drive water influx. For example, at higher pH levels, carboxyl groups ionize into carboxylate anions ($-\text{COO}^-$), increasing electrostatic repulsion among polymer chains and expanding the network. Similarly, amine groups ($-\text{NH}_2$) may protonate in acidic conditions, enabling pH-sensitive swelling that adapts to varying soil environments.

The network architecture, including crosslink density and mesh size (r_{mesh}), plays a critical role in determining both swelling behavior and controlled release. A lower crosslink density creates a more open network structure, allowing greater water absorption, while a higher density restricts swelling due to reduced free volume. Mesh size is particularly important as it determines the diffusional pathways available for water molecules. In networks with larger mesh sizes, water can freely diffuse and interact with polymer chains, resulting in significant swelling. Conversely, smaller mesh sizes impede water uptake, limiting the extent of swelling. Similarly, when the mesh size is significantly larger than the agrochemicals, steric hindrance is negligible, enabling diffusion-dominated release. For agrochemicals larger than the mesh size, steric hindrance leads to immobilization until degradation, swelling, or mechanical deformation occurs [11]. Advanced hydrogel designs, such as double-network systems and hybrid composites, further refine these properties by integrating hydrophilic components with mechanically robust backbones. In particular, combining a soft, hydrophilic polymer network with a rigid, covalently crosslinked structure enables simultaneous enhancement of water retention and mechanical stability.

In agricultural applications, hydrogels often exhibit stimuli-responsive swelling behavior that can also lead to the controlled delivery of agrochemicals. This behavior is triggered by changes in environmental conditions such as pH, temperature, ionic strength, or the presence of specific molecules. For instance, pH-responsive hydrogels rely on the ionization state of functional groups like carboxylic acids or amines. At acidic pH, carboxyl groups remain protonated, minimizing swelling. Conversely, at alkaline pH, these groups deprotonate into carboxylate ions, generating repulsive forces that expand the hydrogel network. Similarly, temperature-responsive hydrogels, often based on polymers such as poly(N-isopropylacrylamide) (PNIPAM), exhibit a lower critical solution temperature (LCST). Below LCST, the network remains hydrophilic and swollen. The swelling induced by changes in temperature and pH facilitates the diffusion of agrochemicals, enabling controlled delivery and targeted

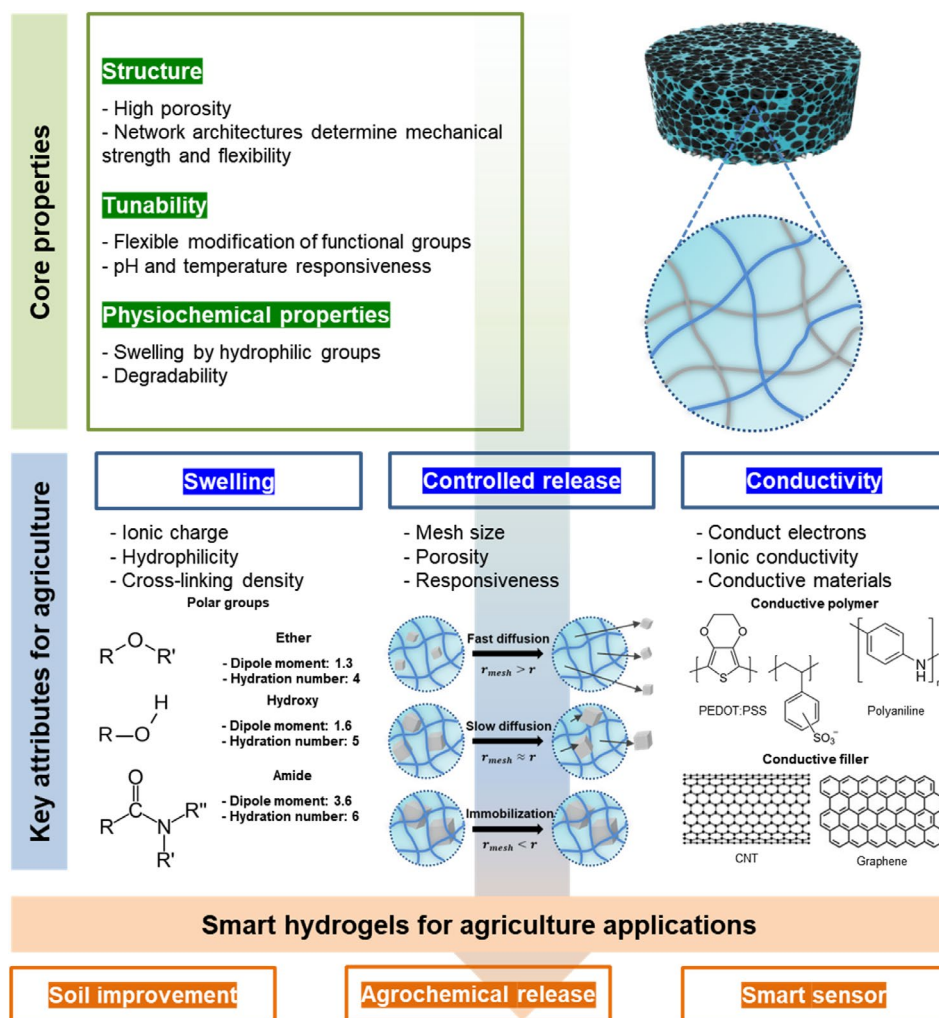


FIGURE 2 | Core properties of hydrogels determined by structure, tunability, and physicochemical characteristics; key attributes include swelling, controlled release, and conductivity for agricultural applications.

release. Ionic strength is another critical trigger, particularly in agricultural soils with varying salinity. High concentrations of dissolved ions screen electrostatic interactions between ionic groups, reducing swelling. This property can be tailored to optimize water retention and nutrient delivery in saline soils. Additionally, hydrogels can be designed to respond to specific agrochemical triggers. Such functional groups that interact with fertilizers or soil nutrients can regulate swelling behavior, enabling controlled release synchronized with crop demand. Incorporating nanoparticles or layered structures such as biochar can further enhance this responsiveness, providing precise control over water and nutrient delivery (Figure 2).

In addition to their swelling and controlled release properties, the hydrophilic nature of hydrogels allows them to dissolve ions and enable ionic conductivity. To increase conductivity, polymers such as polyaniline (PANI), polypyrrole (PPy), or poly(3,4-ethylenedioxythiophene) (PEDOT), as well as conductive nanomaterials like carbon nanotubes (CNTs), graphene, or metal nanoparticles, can be incorporated into hydrogels. In agricultural applications, these conductive hydrogels can function as soil sensors, monitoring environmental parameters in real time. The electrical conductivity of these materials changes in response to environmental stimuli, such as swelling induced by

water absorption, providing measurable signals that reflect soil conditions.

By leveraging these structure–property relationships and responsive chemistries, hydrogels can be effectively utilized in agriculture for multiple purposes. Their ability to adapt to environmental stimuli such as pH, temperature, and ionic strength enables precise and controlled release of agrochemicals. Additionally, advancements in conductive hydrogel technology provide a platform for real-time soil monitoring and the integration of smart farming systems. Consequently, hydrogels represent a pivotal innovation in addressing global agricultural challenges, enhancing resilience to climate variability, and promoting the adoption of eco-friendly practices in modern agriculture.

4 | Integrated Functions of Smart Hydrogels for Agriculture Applications

Essential inputs such as water and agrochemicals can be strategically managed to enhance plant health and productivity. Traditional agricultural practices often rely on the independent application of these inputs, a fragmented approach that leads to

inefficiencies such as water wastage, nutrient leaching, and the accumulation of harmful chemicals in the environment. For instance, over-irrigation can wash away fertilizers before plant uptake, reducing productivity while contaminating groundwater and ecosystems. To address these issues, researchers have developed integrated systems that combine water management with agrochemical delivery. Multifunctional hydrogels provide a dual role in retaining water and controlling the release of nutrients or pesticides. Additionally, conductive hydrogels can be utilized as sensors in agriculture, monitoring critical parameters such as soil moisture, nutrient levels, or environmental conditions. By acting as reservoirs, hydrogels ensure that water and agrochemicals are released in alignment with crop demands, while hydrogel-based sensors provide continuous feedback, enabling farmers to adjust inputs dynamically and further improve resource efficiency.

4.1 | Hydrogel-Based Nutrient Delivery System

The water retention properties of hydrogels, combined with their unique structural characteristics, enable the encapsulation and gradual release of nutrients to plants. As hydrogels can absorb and retain large volumes of water, this trait is particularly valuable in water-scarce regions, as it helps maintain soil moisture and supports plant resilience during dry periods. Controlled fertilizer release through hydrogels predominantly relies on swelling behavior in response to soil moisture. Fertilizers are released through a three-stage process. In the lag phase, water infiltrates the coating, but no significant nutrient release occurs. This is followed by the linear release phase, where nutrients diffuse gradually at a steady rate. Finally, in the decay phase, the release slows as nutrient levels in the hydrogel core drop [12]. Environmental triggers such as pH and temperature often contribute to the slow release. Additionally, hydrogels protect nutrients from rapid leaching, volatilization, and environmental degradation while maintaining steady nutrient availability over time. Such designs incorporate multi-layer coatings, nanoparticle integration, or biochar encapsulation to further enhance release precision, nutrient utilization efficiency, and soil health. Biodegradable materials are commonly used to create these hydrogels, providing environmental compatibility.

Encapsulated biochar-based fertilizers offer controlled nutrient release while improving soil properties, as biochar serves as an effective nutrient carrier that reduces release rates due to its porous structure and abundance [13]. A magnesium-based biochar hydrogel was synthesized by combining magnesium-enriched biochar with polyacrylic acid (PAA), achieving exceptional water absorption of 1395 g g^{-1} and enhanced nutrient release, with approximately 62.1% of nitrogen released over 30 days [14]. Similarly, Wen et al. [15] synthesized microwave-assisted biochar-based slow-release fertilizers by utilizing advanced microwave irradiation techniques [15]. Hydrogen bonding and electrostatic interactions bind ammonium ions onto the biochar, while the hydrogel matrix acts as a physical barrier, enabling gradual nitrogen diffusion. The addition of bentonite creates a more complex nutrient release pathway, and the matrix's biodegradation further extends nutrient availability over time. This approach significantly improved soil water

retention, and cumulative nitrogen release was 69.8% over 30 days. Expanding on the focus on cost-efficiency and eco-friendliness, Das et al. [16] developed a hydrogel-biochar composite using pine resin as a biodegradable polymer backbone [16]. The composite sustained nutrient availability for 45 days in soil while reducing nutrient loss with a biodegradability half-life of 138–162 days, making it ideal for resource-limited agricultural practices. In addition, An et al. introduced a dual system comprising composite biochar derived from cotton straw and montmorillonite and a bio-based hydrophobic biofilm [17]. This combination relies on the properties of biochar as a nutrient carrier and the hydrophobic biofilm as a barrier to control nutrient release. The hydrophobic biofilm slows the release rate by creating a water-resistant layer, ensuring nutrients are gradually available in synchronization with crop demands. The system exhibited a 65% cumulative nutrient release rate over 31 days, effectively matching crop growth needs and demonstrating significant potential for large-scale applications using locally abundant materials.

Multilayer coatings and nanoparticles contribute to advancements in slow-release fertilizer technology. These layered approaches provide a structured, functional design where each layer serves a specific purpose, prolonging the release of nutrients by acting as physical barriers. Nanoparticles, on the other hand, enhance the mechanical strength, nutrient retention, and functional interactions within the hydrogel matrix. A dual-layer coating system, with natural rubber (NR) as the inner layer and NR grafted with PAA as the outer layer, demonstrated controlled nutrient release rates of 54.35% for nitrogen, 51.18% for phosphorus, and 44.37% for potassium over 30 days [18]. The NR layer acts as a hydrophobic barrier, further reducing nutrient leaching into the soil, and the outer layer swells upon contact with soil moisture, controlling the release of nutrients through dynamic diffusion. Noppakundilokrat et al. introduced a trilayer-coated NPK fertilizer hydrogel comprising a polyvinyl alcohol (PVA) coating, a crosslinked chitosan layer, and an outer copolymer hydrogel layer [19]. This system demonstrated a slow-release behavior with 83.9% of nitrogen, 62.3% of phosphorus, and 36.2% of potassium released over 30 days. The interplay of these layers effectively minimized leaching by creating ionic interactions that slowed the diffusion of nutrients like potassium, which tends to bind more strongly to the hydrogel matrix. Salimi et al. [20] developed a system using natural char nanoparticles (NCNPs) as a dual-purpose nanofiller and physical cross-linker to enhance the structural integrity and functionality of the hydrogel matrix [20]. This formulation demonstrated the release of ~70% of nitrogen over 21 days. By integrating nanoparticles with a multilayer coating system, Lu et al. [21] developed a PVA/starch hydrogel membrane enhanced with iron oxide nanoparticles and a biochar outer layer to improve the efficiency and sustainability of nutrient delivery [21]. The hydrogel membrane swells upon absorbing water, forming a diffusion barrier that restricts nutrient release, while the biochar layer adsorbs additional nutrients to further promote gradual diffusion (Figure 3A). The incorporation of iron oxide nanoparticles reduced nitrogen leaching in soil from 31.03% to 24.99% and phosphorus leaching from 37.92% to 36.74% (Figure 3B), while also enhancing the mechanical strength, swelling capacity, and hydrophilicity of the hydrogel membrane, ensuring the gradual and consistent release of essential nutrients.

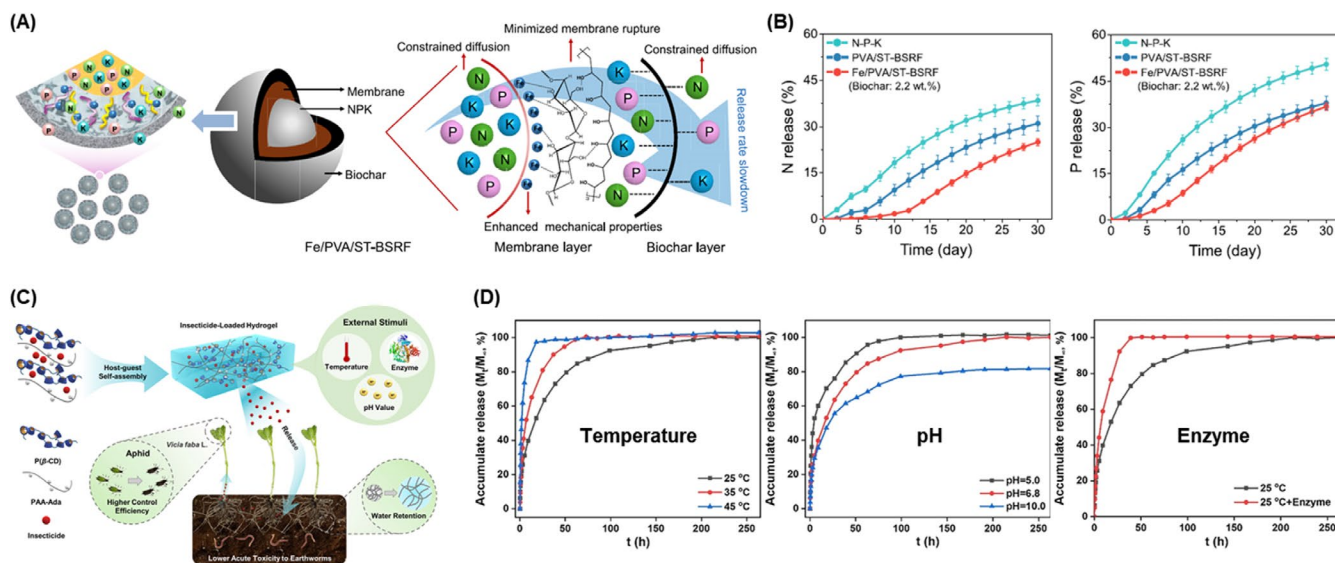


FIGURE 3 | Smart hydrogels for agrochemical release. (A), Schematic diagram of NPK fertilizer release mechanism. (B), release behavior of nitrogen (N) and phosphorus (P) in soil over 30 days. (A, B), Reproduced with permission: Copyright 2024, Elsevier [21]. (C), Schematic diagram of the fabrication of multi-responsive hydrogel system. (D), Release behavior of insecticide under varying conditions, focusing on temperature, pH, and enzymatic action. (C, D), Reproduced with permission: Copyright 2022, American Chemical Society [22].

4.2 | Hydrogel-Based Pesticide Delivery System

Pesticide release through hydrogels leverages stimuli-responsive mechanisms to achieve precision and efficiency in targeting designated areas. These systems are designed to release active ingredients like insecticides or herbicides in response to environmental triggers, such as changes in pH, temperature, or enzymatic activity in pest-infested areas. Similar to fertilizer release, biodegradability is a critical factor, leading to the predominant use of polysaccharides in hydrogel formulations. However, to ensure immediate responsiveness for effective targeting, hydrogels are often integrated with nanomaterials, stimuli-responsive polymers, or chemically grafted functional groups. Unlike fertilizers, which benefit from steady, long-term delivery, pesticides frequently require rapid and targeted action to address pest outbreaks efficiently. This focus on stimuli-responsiveness, inspired by advancements in drug delivery systems, positions hydrogels as a promising platform for sustainable and precise pest control in modern agriculture.

Alginate and polyacrylamide (PAM) are widely used for their biocompatibility, mechanical properties, and hydrophilicity. Singh et al. focused on developing starch-alginate-PAM hydrogels to enhance environmental sustainability by reducing pesticide loss due to photodegradation and leaching [23]. The pesticide release mechanism in their system relies on the swelling properties of the hydrogel, which facilitate a non-Fickian diffusion mechanism that combines diffusion and polymer relaxation. Wang et al., on the other hand, developed hydrophobic alginate-based double-network nanocomposite hydrogels, where hydrophobic modifications slow water uptake and restrict pesticide mobility, ensuring a Fickian diffusion-driven, prolonged release [24]. Nanomaterials, such as soil colloids, incorporated into a hydrogel matrix can act as a barrier to the diffusion of agrochemicals. Soil colloids enhance the hydrogel's

structure by increasing crosslink density, reducing pore size, and creating a more restrictive network. Incorporating natural soil colloids into the interpenetrating network of sodium alginate and PAM hydrogels reduced the pesticide release rate by 40.7% while extending the release duration by 414.3% compared to pure hydrogels [25].

pH responsiveness is critical in pesticide release, ensuring that active compounds are delivered only in specific pH environments, such as acidic soils or plant tissues. The ability of hydrogels to respond to pH changes enables trigger-based mechanisms for controlled pesticide delivery. Under alkaline conditions, functional groups like carboxyl or amine groups within the hydrogel matrix ionize [26, 27], leading to increased water absorption and matrix expansion. This swelling alters the hydrogel's structure, increasing its porosity and facilitating the gradual diffusion of the encapsulated pesticide. Xiang et al. utilized attapulgite (ATP)-based hydrogels combined with polydopamine for the controlled release of chlorpyrifos [28]. ATP served as an environmentally friendly, low-cost carrier that provided a structural base for pesticide adsorption. The system relied on the pH sensitivity of calcium alginate, which swelled and dissolved at higher pH levels, promoting pesticide release. The release rates increased significantly at higher pH levels (from 60% at pH 5.5 to 92% at pH 8.5), demonstrating effective delivery based on environmental conditions. Similarly, Zha et al. utilized dopamine-modified ATP combined with sodium alginate to achieve pH-dependent release of glyphosate while also enhancing its stability against environmental factors such as UV light and temperature [29]. This system exploited the ionization of carboxyl groups in sodium alginate, which caused the microspheres to swell and release glyphosate in response to alkaline conditions. The study reported a 60% release of glyphosate at pH 8.5 within 24 h, compared to only 20% under neutral or acidic pH. Xiang et al. used biochar for dual-stimuli responsive system that can respond to two environmental factors, pH

and ionic strength [30]. At low pH, the system swells, enhancing diffusion-based release. Also, increased ionic strength disrupts the hydrogel's compact structure, facilitating pesticide release via ion-exchange and diffusion processes. The hydrogel's porous structure and electrostatic interactions allow gradual and sustained pesticide release, achieving 95% of the pesticide released within 72 h at pH 7.0. While swelling behavior is a primary trigger for controlled release in many systems, some pH-responsive mechanisms operate independently or rely minimally on swelling. For example, hydrazone bonds are pH-responsive, breaking down in acidic conditions while remaining stable in neutral and alkaline environments [31]. Similarly, gradual hydrolysis of ester bonds and systems based on hydrogen bonding [32] and ionic interactions can be designed to respond to pH and ionic strength without significant matrix swelling [33].

Along with pH responsiveness, other stimuli-responsive materials are incorporated to enhance the versatility, precision, and effectiveness of agrochemical delivery systems in diverse scenarios. By integrating additional responsiveness mechanisms such as redox, temperature, and light, these systems can be finely tuned to provide precise release in response to specific environmental triggers and ensure agrochemicals are activated only in target environments. PNIPAM is widely employed for its phase-transition capability, utilizing its inherent LCST. For instance, Gu et al. synthesized a double-network hydrogel system using dehydroabietic acid derived from rosin and PNIPAM as the primary network, with an ionically crosslinked dynamic secondary network formed by zinc ions [34]. This design achieves both pH and temperature responsiveness, enabling the controlled release of the pesticide avermectin. Similarly, Zheng et al. incorporated the photothermal material semi-coke to introduce light responsiveness, allowing the generation of heat to induce PNIPAM's phase transition [35]. They developed a multi-responsive grafted copolymer hydrogel, enhancing versatility under light, temperature, and pH conditions. Another multi-responsive hydrogel system was proposed by Wu et al., which combined pH, temperature, and enzymatic activity responsiveness (Figure 3C) [22]. This system was formed using poly(β -cyclodextrin) and PAA-adamantane. Enzymatic activity triggered by amyloglucosidase hydrolyzes β -cyclodextrin, breaking down the hydrogel structure and causing a rapid burst release. While this system does not incorporate a traditional thermoresponsive polymer, it achieves temperature responsiveness via hydrogen bond modulation in PAA and the thermal sensitivity of β -cyclodextrin and adamantane host-guest complexes. This multi-responsive hydrogel system showed adaptive release performance, with faster insecticide release at 45°C (80% in 120 h) compared to 25°C (200 h). In acidic conditions (pH 5.0), nearly 100% release occurs in 180 h, while alkaline conditions (pH 10.0) slow-release to 80%. Enzymatic activity accelerates release, completing within 50 h, making the system efficient and responsive to diverse environmental conditions (Figure 3D). Furthermore, leveraging natural redox conditions in plant tissues and soils offers significant potential for precision delivery. Hou et al. [36] prepared a redox- and pH-responsive nanogel composed of carboxymethyl cellulose as the biodegradable and hydrophilic backbone, disulfide bonds for redox responsiveness triggered by reducing agents such as glutathione, and palmitic chloride to introduce hydrophobic branches [36]. This configuration enhanced the loading

capacity and stability of lipid-soluble agrochemicals, such as salicylic acid.

4.3 | Hydrogel-Based Self-Irrigation System

Previous research predominantly highlights the role of hydrogels in soil improvement through their water retention properties. However, substantial benefits could be realized if these hydrogels could directly release the absorbed water into the soil upon varying environmental conditions. This capability requires overcoming inherent limitations. Hydrogels must exhibit both high water absorption capacity and responsive functional groups to facilitate phase changes. Thermoresponsive polymers, which alter their properties in response to temperature fluctuations, present a promising solution. As previously mentioned, PNIPAM is a widely studied material, owing to its well-defined LCST around 32°C. Below the LCST, PNIPAM exists in a hydrated, hydrophilic state, where water molecules surround and stabilize the polymer's isopropyl groups, maintaining solubility. Above this temperature, PNIPAM undergoes a reversible phase transition, where increased hydrophobic interactions among isopropyl groups displace water molecules and weaken amide-water hydrogen bonds, leading to polymer-polymer interactions that drive efficient phase separation [37]. These temperature-responsive characteristics of PNIPAM could be strategically adapted to agricultural applications, enabling water release directly into the soil as temperatures shift.

Adding to the thermoresponsive capabilities of PNIPAM, additional strategies have been developed to mitigate water scarcity challenges, especially in arid regions. For instance, Zhou et al. developed a super moisture-absorbent gel (SMAG) for atmospheric water harvesting [38]. This hybrid gel integrates hygroscopic chloride-doped PPy with PNIPAM's switchable hydrophilicity, enabling efficient water vapor capture and release across a broad range of humidity (Figure 4A). This SMAG-soil system is powered by solar energy to capture atmospheric moisture and release it as liquid water upon heating, providing a sustainable irrigation solution independent of traditional water sources [40]. Capable of maintaining high relative humidity levels within planting chambers, SMAG-soil supports seed germination and plant growth in drought-prone areas, reducing reliance on scarce water resources. Field tests demonstrated that SMAG can sustain plants for 14 days without additional water, achieving a 95% germination rate, substantially higher than that in conventional sandy soil.

Building upon the integration of advanced hydrogels for water absorption and release, a recent research has combined these properties with nutrient delivery systems. Park et al. introduced a self-irrigation and slow-release fertilizer hydrogel (SISRH) system designed to improve water and nutrient efficiency in agriculture [39]. This hydrogel employs a thermoresponsive interpenetrating network incorporating calcium chloride as both a hygroscopic agent and a nutrient source, enabling it to harvest moisture at night and release water and nutrients during the day (Figure 4B). SISRH incorporated in soil showed higher survival rates compared to regular soil while reducing water

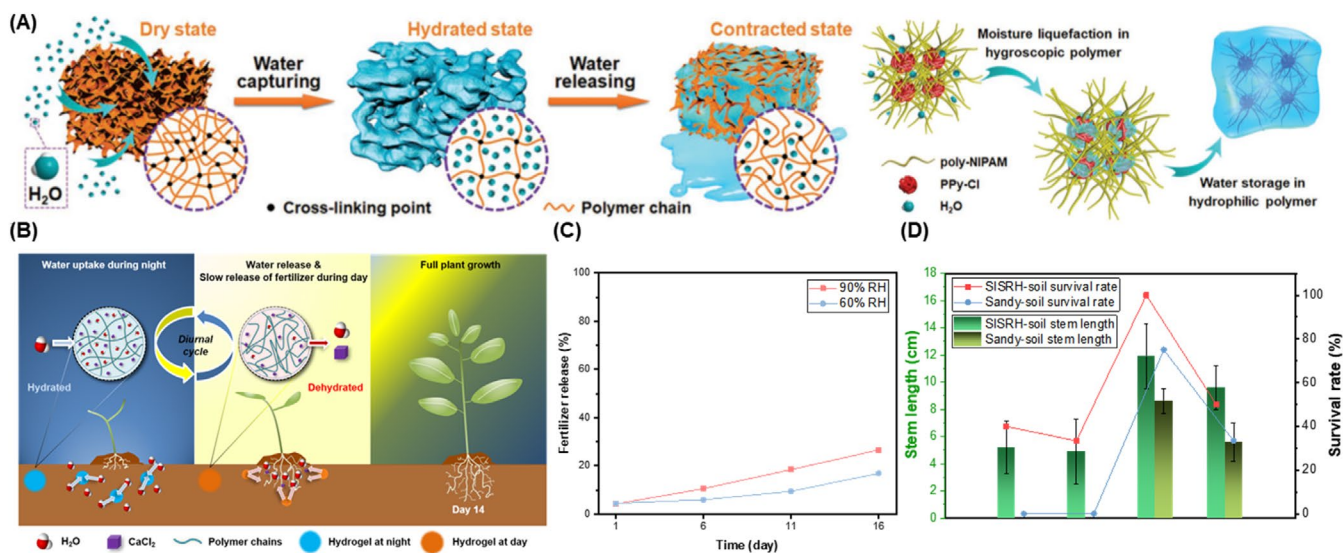


FIGURE 4 | Smart hydrogels for self-irrigation. (A), Schematic illustration of water capturing/releasing and moisture absorption by SMAG. Reproduced with permission: Copyright 2019, Wiley-VCH GmbH [38]. (B), Schematic diagram of water absorption/release and fertilizer release operating in a diurnal cycle for SISRH. (C), SISRH performance for fertilizer release in different RH conditions. (D), Comparison of SISRH-soil and sandy-soil in plant growth and survival rate. (B–D), Reproduced with permission: Copyright 2024, American Chemical Society [39].

consumption by ~40%. SISRH optimizes plant growth through sustained nutrient release, significantly improving crop yield and resilience in water-limited environments (Figure 4C,D). This approach mitigates critical challenges in water efficiency and soil fertility, offering a promising solution for sustainable agriculture under climate-stressed conditions.

Another recent advancement involves the solar-driven scalable hygroscopic gel for water recycling and efficient irrigation [41]. This gel incorporates titanium nitride for solar photothermal energy absorption, curdlan for structural stability, and sodium polyacrylate (PAAS) with lithium chloride to enhance hygroscopicity. Functioning through a diurnal cycle, the gel captures water vapor at night and releases it as liquid during the day by solar energy. This system minimizes irrigation needs, achieving water savings of ~44.9%.

4.4 | Hydrogel-Based Sensor System

Hydrogel-based smart sensors are transforming agriculture by offering advanced tools for precision farming. These sensors enable real-time monitoring of key environmental and plant physiological parameters. Incorporating conductive polymers facilitates electron transfer and signal transduction, broadening their applications in environmental monitoring and precision agriculture. By integrating functionalities such as water management, energy harvesting, multifunctional sensing, and environmental contamination detection, hydrogel-based sensors meet the growing demands for sustainable and high-yield agricultural practices. Their adaptability to diverse agricultural conditions, combined with cost-effective and scalable fabrication methods, establishes them as essential technologies for addressing food security and environmental challenges in modern farming.

The integration of PEDOT:poly(styrene sulfonate) (PEDOT:PSS) into hydrogel-based systems enhances their electrical conductivity

and facilitates the conversion of ionic to electronic signals, enabling efficient transduction for real-time monitoring of environmental parameters [42–44]. A conductive hydrogel incorporating PEDOT:PSS can function as a solenoid valve controller, utilizing its moisture-conductivity relationship to regulate irrigation based on soil moisture levels [45]. This approach addresses the water inefficiency in agriculture by enabling intelligent irrigation systems. Guo et al. [46] introduced a multifunctional hydrogel that integrates energy harvesting and self-powered sensing capabilities in a single device for real-time, noninvasive monitoring of leaf water content [46]. The ionic hydrogel, composed of PVA/PAM with lithium chloride and glycerol, enables ionic conductivity by facilitating the movement of Li^+ and Cl^- ions. The system achieved a stable direct current output with an average power density of 1.9 W m^{-3} and demonstrates self-recovery in high temperatures and low humidity environmental conditions. Similarly, a self-powered triboelectric energy harvester (TEG) using reduced graphene oxide as a conductive nanofiller dispersed throughout the polyacrylic acid hydrogel, with a conductive coating of PANI, was developed [47]. This device achieved a high-power density of 424 mW m^{-2} by harvesting energy from acoustic, mechanical, and wind sources. It demonstrated superior stretchability and sensitivity, enabling it to monitor plant growth and detect ammonia gas concentrations as low as 8.62 ppm within 60 s. The hydrogel-powered LED treatment resulted in an average increase in leaf length of 35% (Figure 5B). Wang et al. developed an environmental contamination detection device using a portable and flexible hydrogel sensor to detect atrazine (ATZ) residues (Figure 5C) [48]. The device integrates graphite-phase carbon nitride ($\text{g-C}_3\text{N}_4$) into an aramid nanofiber-reinforced PVA hydrogel, enabling rapid, sensitive, and on-site detection of ATZ with a limit of detection as low as 0.002 ng mL^{-1} . The hydrogel's high conductivity and mechanical strength is from the hybrid structure of $\text{g-C}_3\text{N}_4$ and PVA, with aramid nanofibers providing robustness. The hydrogel sensor's performance aligned closely with high-performance liquid chromatography (HPLC) in detecting ATZ residues in lake water, farmland soil, and agricultural products, enabling reliable on-site

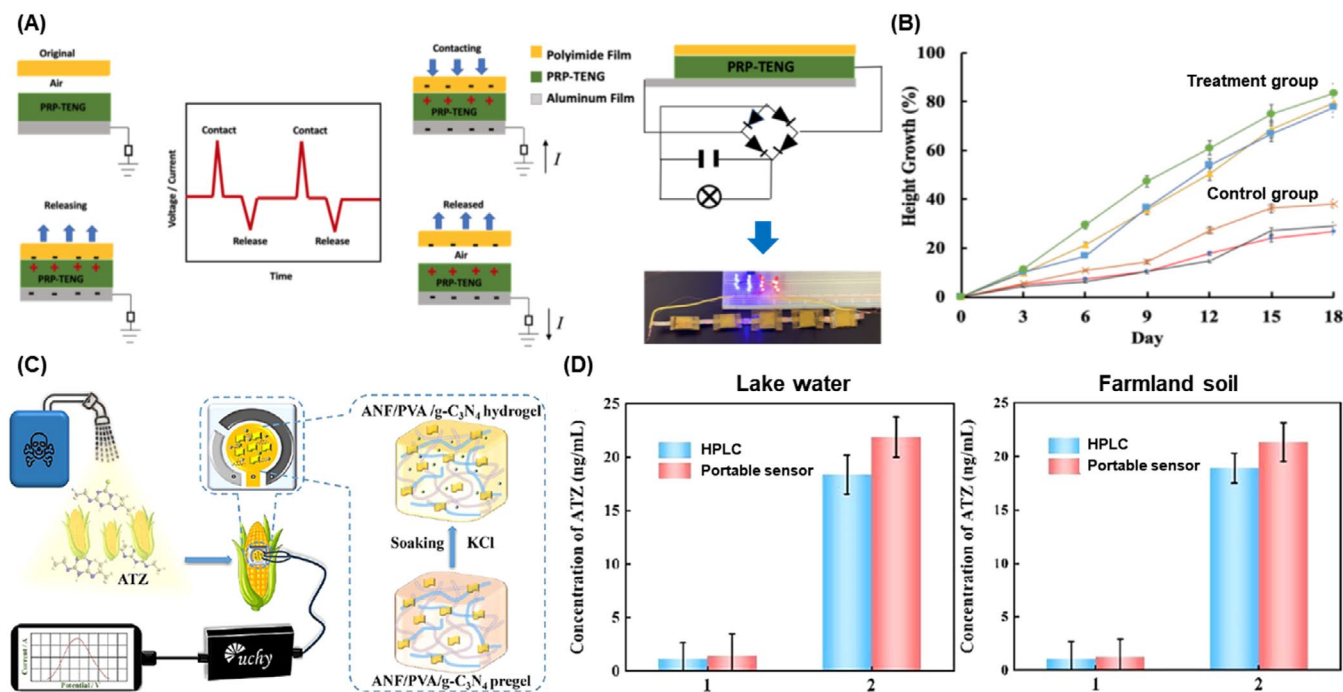


FIGURE 5 | Smart hydrogel-based sensor. (A), Energy conversion process of hydrogel-based TENG through cyclic contact-release and application in LED powering. (B), Growth rate differences in the control group and LED treatment group. (B, C), Reproduced with permission: Copyright 2021, Elsevier [47]. (C), Schematic illustration of the portable hydrogel electrode for ATZ detection. (D), Comparison of ATZ residues in lake water and farmland soil by HPLC and hydrogel-based portable sensor. (C, D), Reproduced with permission: Copyright 2024, American Chemical Society [48].

analysis with detection times under 1 min and recovery rates of 93%–109% (Figure 5D).

5 | Current Challenges and Outlook

Agricultural systems face critical challenges in optimizing resource utilization, minimizing environmental impact, and ensuring sustainable productivity. While slow-release fertilizers, controlled pesticide delivery systems, and hydrogel-based smart sensors individually address specific issues, their combined limitations underscore the need for integrated solutions. For instance, slow-release fertilizers may not always align nutrient release with plant uptake dynamics, leading to inefficiencies such as nutrient leaching or depletion during peak demand. Similarly, controlled pesticide delivery systems face challenges in maintaining sustained, site-specific release, increasing the risks of pest resistance and non-target effects. Hydrogel-based smart sensors, though capable of real-time monitoring, often suffer from limited mechanical durability, slow response times, and vulnerability to environmental stressors. These issues highlight the need for a multifunctional hydrogel system that integrates and balances diverse functionalities to support sustainable agriculture.

An all-in-one hydrogel system can integrate precise stimuli-responsive agrochemical release mechanisms with embedded sensing capabilities (Figure 6). A polymeric network with tailored porosity and functional groups could dynamically respond to environmental cues, such as changes in soil moisture, temperature, or pH, modulating agrochemical release with high precision and monitoring functionality. PNIPAM is a promising thermoresponsive material with a tunable LCST, contracting at high temperatures to reduce water content and

slow agrochemical diffusion through contraction and osmotic pressure. Alternatively, hydroxypropyl cellulose (HPC) can also be considered particularly due to its biodegradability as a cellulose ether derivative [49, 50]. Compared to PNIPAM, HPC typically has a higher LCST, making it more suitable for applications in high-temperature regions where a delayed or sustained response is required. In addition to thermoresponsive hydrogels, pH-responsive hydrogels such as sodium alginate can be utilized due to the ionization of carboxyl or amine groups within the hydrogel matrix under specific pH conditions [51, 52]. This ionization induces swelling or deswelling behavior, altering the hydrogel's permeability, mechanical properties, and diffusion characteristics. Both thermoresponsive and pH-responsive characteristics can be realized through a cross-linked polymer network [53] or a double network, while also achieving high conductivity [54]. However, the interaction between the two responsive components must be synergistic, as swelling due to pH changes can affect the thermal response. Additionally, it is important to consider whether the response occurs sequentially or simultaneously. Crosslinking strategies should be evaluated to determine whether an IPN, semi-IPN, or hybrid network is the most appropriate choice. The architectural design of the hydrogel matrix is crucial for controlled release applications. Achieving an optimal balance in crosslinking density is essential, as lower crosslinking density enhances water retention but also results in a larger mesh size, which can lead to faster release kinetics. To regulate both water retention and the slow diffusion of agrochemicals, multilayer coatings can be employed to create a core-shell structure that acts as a barrier to modulate release. Hydrophilic or hydrophobic polymers, such as PVA or polylactic acid, can be coated onto the core hydrogel to fine-tune release properties and enable stepwise release in response to fluctuating environmental conditions. However, optimizing

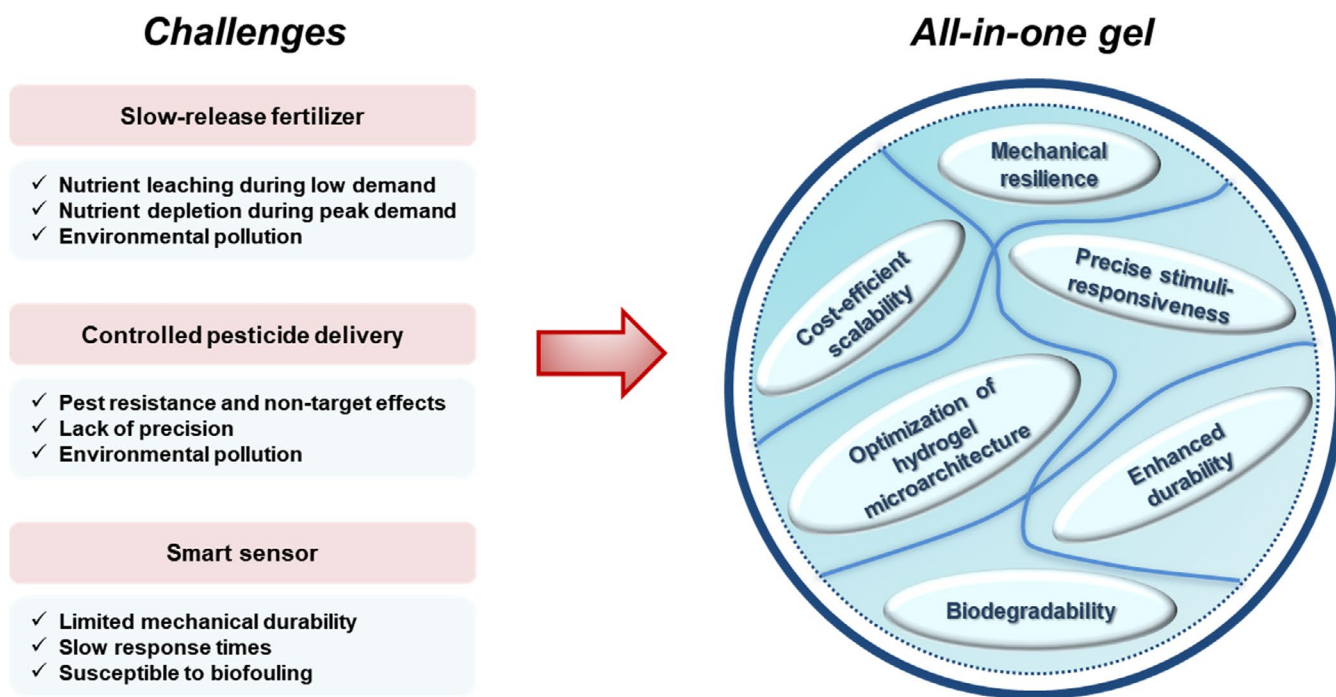


FIGURE 6 | Challenges of current hydrogel-based agriculture technologies and potential solution of all-in-one gel.

the microarchitecture of hydrogels is crucial, as enhancing their agrochemical delivery capabilities may compromise the performance of integrated sensor components. The barrier thickness may be important for hydrogel-based sensor systems as it can influence the sensitivity and response time of the sensor by controlling the diffusion rate of analytes or environmental signals to the sensor surface. Additionally, a well-designed barrier thickness is vital for protecting the sensor from environmental degradation, ensuring its long-term functionality. However, for nutrient delivery, the thicker the barrier, the more difficult it becomes for the diffusion of nutrients to occur. Furthermore, the release of agrochemicals in response to environmental stimuli might be significantly hindered due to delayed or incomplete penetration of triggering agents such as moisture, pH changes, or temperature variations.

Environmental sustainability is integral to this approach. Using biodegradable polymers, such as chitosan or cellulose derivatives, ensures that the hydrogel decomposes into non-toxic byproducts, improving soil health and preventing microplastic pollution. To enhance durability in agricultural conditions, mechanically robust hydrogels reinforced with antimicrobial agents or crosslinkers could withstand environmental stresses over multiple growing seasons. Additionally, incorporating eco-friendly additives, such as biochar or natural clay nanoparticles, could improve nutrient retention and soil conditioning properties. From an economic perspective, the initial production costs of polysaccharide-based hydrogels can be higher than synthetic polyacrylate hydrogels due to the additional chemical treatments required for extraction and processing [55]. However, biodegradable hydrogels offer long-term cost advantages by naturally decomposing and preventing soil contamination. Furthermore, sourcing natural polymers from agricultural waste, such as cellulose from crop residues, can reduce raw material expenses, improving the cost-effectiveness of biodegradable hydrogels over time. Therefore, a critical analysis of the trade-offs between the

water retention properties of biodegradable polymers and the environmental responsiveness of synthetic polymers is essential for economic feasibility in field-scale applications. Ensuring scalability through cost-effective manufacturing with accessible materials is essential to make this technology viable for farmers, particularly in resource-limited regions.

By integrating slow-release fertilizers, controlled pesticide delivery, and smart sensing technologies, this multifunctional hydrogel system offers a comprehensive solution to major agricultural challenges. It enhances resource efficiency, increases crop yields, and reduces environmental impact. Achieving its full potential will require advancing material innovations, conducting extensive field trials across diverse climates, and establishing supportive policies to enable widespread adoption. This approach could redefine sustainable agriculture and make a significant contribution to global food security.

Author Contributions

J.P.: draft preparation and revision. W.G.: review and editing. G.Y.: supervision.

Acknowledgments

G.Y. acknowledges the financial support from the Welch Foundation Award F-1861, the Norman Hackerman Award in Chemical Research, and the Camille-Dreyfus Teacher-Scholar Award.

Conflicts of Interest

The authors declare no conflicts of interest.

References

1. United Nations, "Global Issues," <https://www.un.org/en/global-issues/population>.

2. M. van Dijk, T. Morley, M. L. Rau, and Y. Saghai, "A Meta-Analysis of Projected Global Food Demand and Population at Risk of Hunger for the Period 2010–2050," *Nature Food* 2 (2021): 494–501.
3. P. Greve, T. Kahil, J. Mochizuki, et al., "Global Assessment of Water Challenges Under Uncertainty in Water Scarcity Projections," *Nature Sustainability* 1 (2018): 486–494.
4. E. R. Jones, M. F. P. Bierkens, and M. T. H. van Vliet, "Current and Future Global Water Scarcity Intensifies When Accounting for Surface Water Quality," *Nature Climate Change* 14 (2024): 629–635.
5. H. Mansouri, H. Ait Said, H. Noukrati, A. Oukarroum, H. Ben Youcef, and F. Perreault, "Advances in Controlled Release Fertilizers: Cost-Effective Coating Techniques and Smart Stimuli-Responsive Hydrogels," *Advanced Sustainable Systems* 7 (2023): 2300149.
6. Y. Guo and G. Yu, "Materials Innovation for Global Water Sustainability," *ACS Materials Letters* 4 (2022): 713–714.
7. W. Shi, W. Guan, C. Lei, and G. Yu, "Sorbents for Atmospheric Water Harvesting: From Design Principles to Applications," *Angewandte Chemie, International Edition* 61 (2022): e202211267.
8. C. Lei, W. Guan, Y. Zhao, and G. Yu, "Chemistries and Materials for Atmospheric Water Harvesting," *Chemical Society Reviews* 53 (2024): 7328–7362.
9. X. Zhou, Y. Guo, F. Zhao, and G. Yu, "Hydrogels as an Emerging Material Platform for Solar Water Purification," *Accounts of Chemical Research* 52 (2019): 3244–3253.
10. Y. Guo, J. Bae, Z. Fang, P. Li, F. Zhao, and G. Yu, "Hydrogels and Hydrogel-Derived Materials for Energy and Water Sustainability," *Chemical Reviews* 120 (2020): 7642–7707.
11. J. Li and D. J. Mooney, "Designing Hydrogels for Controlled Drug Delivery," *Nature Reviews Materials* 1 (2016): 16071.
12. E. Priya, S. Sarkar, and P. K. Maji, "A Review on Slow-Release Fertilizer: Nutrient Release Mechanism and Agricultural Sustainability," *Journal of Environmental Chemical Engineering* 12 (2024): 113211.
13. D. H. H. Sim, I. A. W. Tan, L. L. P. Lim, and B. H. Hameed, "Encapsulated Biochar-Based Sustained Release Fertilizer for Precision Agriculture: A Review," *Journal of Cleaner Production* 303 (2021): 127018.
14. Z. Lang, S. Yan, and Q. Zhu, "Water Retention and Sustained Release of Magnesium-Based Biochar Modified Hydrogel Composite Materials," *Journal of Environmental Chemical Engineering* 11 (2023): 111380.
15. P. Wen, Z. Wu, Y. Han, G. Cravotto, J. Wang, and B. C. Ye, "Microwave-Assisted Synthesis of a Novel Biochar-Based Slow-Release Nitrogen Fertilizer With Enhanced Water-Retention Capacity," *ACS Sustainable Chemistry & Engineering* 5 (2017): 7374–7382.
16. S. K. Das and G. K. Ghosh, "Hydrogel-Biochar Composite for Agricultural Applications and Controlled Release Fertilizer: A Step Towards Pollution-Free Environment," *Energy* 242 (2022): 122977.
17. T. An, H. Cheng, Y. Qin, et al., "The Dual Mechanisms of Composite Biochar and Biofilm Towards Sustainable Nutrient Release Control of Phosphate Fertilizer: Effect on Phosphorus Utilization and Crop Growth," *Journal of Cleaner Production* 311 (2021): 127329.
18. Y. Cui, Y. Xiang, Y. Xu, et al., "Poly-Acrylic Acid Grafted Natural Rubber for Multi-Coated Slow Release Compound Fertilizer: Preparation, Properties and Slow-Release Characteristics," *International Journal of Biological Macromolecules* 146 (2020): 540–548.
19. S. Noppakundilongrat, N. Phecharat, and S. Kiatkamjornwong, "Multilayer-Coated NPK Compound Fertilizer Hydrogel With Controlled Nutrient Release and Water Absorbency," *Journal of Applied Polymer Science* 132 (2015): 41249.
20. M. Salimi, E. Motamedi, B. Moteszarezedeh, H. M. Hosseini, and H. A. Alikhani, "Starch-g-Poly(Acrylic Acid-Co-Acrylamide) Composites Reinforced With Natural Char Nanoparticles Toward Environmentally Benign Slow-Release Urea Fertilizers," *Journal of Environmental Chemical Engineering* 8 (2020): 103765.
21. J. Lu, M. Wu, L. Luo, et al., "Incorporating Iron Oxide Nanoparticles in Polyvinyl Alcohol/Starch Hydrogel Membrane With Biochar for Enhanced Slow-Release Properties of Compound Fertilizers," *Carbohydrate Polymers* 348 (2025): 122834.
22. T. Wu, K. Zhao, C. Zhang, et al., "Promising Delivery Platform for Smart Pest Control With High Water-Retaining Capacity," *ACS Applied Materials & Interfaces* 14 (2022): 55062–55074.
23. J. Singh, B. Singh, and Vishavnath, "Designing Starch–Alginate Hydrogels for Controlled Delivery of Fungicide for the Alleviation of Environmental Pollution," *ACS Agricultural Science & Technology* 2 (2022): 1239–1250.
24. L. Wang, G. Yu, J. Li, et al., "Stretchable Hydrophobic Modified Alginate Double-Network Nanocomposite Hydrogels for Sustained Release of Water-Insoluble Pesticides," *Journal of Cleaner Production* 226 (2019): 122–132.
25. Q. Zhang, G. Yu, Q. Zhou, et al., "Eco-Friendly Interpenetrating Network Hydrogels Integrated With Natural Soil Colloid as a Green and Sustainable Modifier for Slow Release of Agrochemicals," *Journal of Cleaner Production* 269 (2020): 122060.
26. R. Zhang, M. Tang, A. Bowyer, R. Eissenthal, and J. Hubble, "A Novel pH- and Ionic-Strength-Sensitive Carboxymethyl Dextran Hydrogel," *Biomaterials* 26 (2005): 4677–4683.
27. V. Kozlovskaya, E. Kharlampieva, M. L. Mansfield, and S. A. Sukhishvili, "Poly(Methacrylic Acid) Hydrogel Films and Capsules: Response to pH and Ionic Strength, and Encapsulation of Macromolecules," *Chemistry of Materials* 18 (2006): 328–336.
28. Y. Xiang, G. Zhang, C. Chen, B. Liu, D. Cai, and Z. Wu, "Fabrication of a pH-Responsively Controlled-Release Pesticide Using an Attapulgit-Based Hydrogel," *ACS Sustainable Chemistry & Engineering* 6 (2018): 1192–1201.
29. X. Zha, X. Hou, Q. Li, et al., "Loading Glyphosate in Attapulgit and Sodium Alginate Hydrogels to Construct pH-Responsive Controlled Release Microspheres for Enhanced Soil Sustained Release," *ACS Agricultural Science & Technology* 2 (2022): 1090–1100.
30. Y. Xiang, X. Lu, J. Yue, et al., "Stimuli-Responsive Hydrogel as Carrier for Controlling the Release and Leaching Behavior of Hydrophilic Pesticide," *Science of the Total Environment* 722 (2020): 137811.
31. G. Liu, G. Lin, M. Tan, et al., "Hydrazone-Linked Soybean Protein Isolate-Carboxymethyl Cellulose Conjugates for pH-Responsive Controlled Release of Pesticides," *Polymer Journal* 51 (2019): 1211–1222.
32. L. Zheng, F. Seidi, W. Wu, Y. Pan, and H. Xiao, "Dual-Functional Lignin-Based Hydrogels for Sustained Release of Agrochemicals and Heavy Metal Ion Complexation," *International Journal of Biological Macromolecules* 235 (2023): 123701.
33. C. Xu, L. Cao, C. Cao, et al., "Fungicide Itself as a Trigger to Facilely Construct Hymexazol-Encapsulated Polysaccharide Supramolecular Hydrogels With Controllable Rheological Properties and Reduced Environmental Risks," *Chemical Engineering Journal* 452 (2023): 139195.
34. S. Gu, M. Liu, R. Xu, et al., "Ecofriendly Controlled-Release Insecticide Carrier: pH-/Temperature-Responsive Rosin-Derived Hydrogels for Avermectin Delivery Against Mythimna Separata (Walker)," *Langmuir* 40 (2024): 10992–11010.
35. D. Zheng, K. Wang, B. Bai, N. Hu, and H. Wang, "Swelling and Glyphosate-Controlled Release Behavior of Multi-Responsive Alginate-g-P(NIPAm-Co-NDEAm)-Based Hydrogel," *Carbohydrate Polymers* 282 (2022): 119113.
36. X. Hou, Y. Pan, H. Xiao, and J. Liu, "Controlled Release of Agrochemicals Using pH and Redox Dual-Responsive Cellulose Nanogels," *Journal of Agricultural and Food Chemistry* 67 (2019): 6700–6707.

37. W. Guan, Y. Zhao, C. Lei, and G. Yu, "Molecularly Confined Hydration in Thermoresponsive Hydrogels for Efficient Atmospheric Water Harvesting," *Proceedings of the National Academy of Sciences of the United States of America* 120 (2023): e2308969120.
38. F. Zhao, X. Zhou, Y. Liu, Y. Shi, Y. Dai, and G. Yu, "Super Moisture-Absorbent Gels for All-Weather Atmospheric Water Harvesting," *Advanced Materials* 31 (2019): 1806446.
39. J. Park, W. Guan, C. Lei, and G. Yu, "Self-Irrigation and Slow-Release Fertilizer Hydrogels for Sustainable Agriculture," *ACS Materials Letters* 6 (2024): 3471–3477.
40. X. Zhou, P. Zhang, F. Zhao, and G. Yu, "Super Moisture Absorbent Gels for Sustainable Agriculture via Atmospheric Water Irrigation," *ACS Materials Letters* 2 (2020): 1419–1422.
41. H. Zou, X. Yang, J. Zhu, et al., "Solar-Driven Scalable Hygroscopic Gel for Recycling Water From Passive Plant Transpiration and Soil Evaporation," *Nature Water* 2 (2024): 663–673.
42. L. V. Kayser and D. J. Lipomi, "Stretchable Conductive Polymers and Composites Based on PEDOT and PEDOT:PSS," *Advanced Materials* 31 (2019): 1806133.
43. B. Lu, H. Yuk, S. Lin, et al., "Pure PEDOT:PSS Hydrogels With High Electrical Conductivity, Stretchability, and Stability for Bioelectronics," *Nature Communications* 10 (2019): 1043.
44. C. Sun, X. Wang, M. A. Auwalu, S. Cheng, and W. Hu, "Organic Thin Film Transistors-Based Biosensors," *EcoMat* 3 (2021): e12094.
45. X. Q. Zhan, Z. Q. Ran, H. Y. Bao, et al., "Intelligent Hydrogel On-Off Controller Sensor for Irrigation," *Advanced Composites and Hybrid Materials* 7 (2024): 6.
46. X. Guo, L. Wang, Z. Jin, and C. Lee, "A Multifunctional Hydrogel With Multimodal Self-Powered Sensing Capability and Stable Direct Current Output for Outdoor Plant Monitoring Systems," *Nano Micro Letters* 17 (2025): 76.
47. H. H. Hsu, X. Zhang, K. Xu, et al., "Self-Powered and Plant-Wearable Hydrogel as LED Power Supply and Sensor for Promoting and Monitoring Plant Growth in Smart Farming," *Chemical Engineering Journal* 422 (2021): 129499.
48. Y. Wang, Y. Wang, Y. Xue, et al., "Portable and Flexible Hydrogel Sensor for On-Site Atrazine Assay on Agricultural Products," *Analytical Chemistry* 96 (2024): 7772–7779.
49. M. Gosecki, H. Setälä, T. Virtanen, and A. J. Ryan, "A Facile Method to Control the Phase Behavior of Hydroxypropyl Cellulose," *Carbohydrate Polymers* 252 (2021): 117015.
50. J. Gao, G. Haidar, X. Lu, and Z. Hu, "Self-Association of Hydroxypropylcellulose in Water," *Macromolecules* 34 (2001): 2242–2247.
51. Y. Li, C. Wang, Y. Luan, et al., "Preparation of pH-Responsive Cellulose Nanofibril/Sodium Alginate Based Hydrogels for Drug Release," *Journal of Applied Polymer Science* 139 (2022): e51647.
52. M. A. Abd El-Ghaffar, M. S. Hashem, M. K. El-Awady, and A. M. Rabie, "pH-Sensitive Sodium Alginate Hydrogels for Riboflavin Controlled Release," *Carbohydrate Polymers* 89 (2012): 667–675.
53. Y. Bai, Z. Zhang, A. Zhang, et al., "Novel Thermo- and pH-Responsive Hydroxypropyl Cellulose- and Poly (L-Glutamic Acid)-Based Microgels for Oral Insulin Controlled Release," *Carbohydrate Polymers* 89 (2012): 1207–1214.
54. S. Gan, S. Bai, C. Chen, et al., "Hydroxypropyl Cellulose Enhanced Ionic Conductive Double-Network Hydrogels," *International Journal of Biological Macromolecules* 181 (2021): 418–425.
55. A. Riofrio, T. Alcivar, and H. Baykara, "Environmental and Economic Viability of Chitosan Production in Guayas-Ecuador: A Robust Investment and Life Cycle Analysis," *ACS Omega* 6 (2021): 23038–23051.